

Video Glaze: Anti-Style Mimicry Software for Animation

Team Name: (deep) fake it till you make it

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Overview

AI generated videos have become increasingly popular in recent years, including videos of fake actors, and AI-generated animation [2,3]. There are a number of tools that have been proposed to cloak real photos and videos from deepfakes [4,5], but few to prevent stylistic mimicry for art, and only for images [1]. Our goal is to apply a pre-existing tool for anti-AI style mimicry in images to videos. Users of this novel tool could apply it to their own animations and art videos to protect them from AI generation tools. Alongside further researching privacy preservation tools for AI imaging and text generation, we hope to empirically test our topic by experimenting with anti-AI image tools on the market.

To explore this topic further, we are looking to apply the anti-AI image generation tool Glaze to combat AI art video generation [1]. Specifically, we would like to explore if different video generation models respond differently to defensive perturbations. We would like to evaluate whether Glaze-based protections are consistently effective across multiple systems or maybe whether some models are able to bypass or ignore these perturbations. Our goal is to assess the robustness and transferability of defensive techniques in the video generation area. Although Glaze cannot be used retroactively on images that have already been uploaded to the internet, it can be used proactively moving forward for new images that people will post online, and we are seeking to extend this future protection to videos.

We will use a combination of open source animation clips and one self-created animation as the target styles we are looking to “protect.” We will focus on more recent clips that were likely not used to train the generator models, and include a homemade clip that could not possibly have been included in training as a control. We will apply Glaze to the video data by protecting all the frames. After the defensive measures are put in place, we will evaluate 3 text-to-video systems: Runway Gen-2 (commercial product), Pika Labs (stylization focused model), and ModelScope text-to-video (research based, open source). For each of the models, we provide prompts to replicate the style of the original video and generate outputs using the original videos and Glaze-protected videos. In the end, we will compare how well each model reproduces the original style under both conditions. Our evaluation metrics include a quantitative approach by analyzing things like frame level similarity, a style mimicry analysis that measures how closely the output videos match the original and a more general human evaluation that may include a user study to gather their opinions.

Research Questions

1. What are current video animation style protection measures and what does our method (applying Glaze frame-by-frame) add?
2. How effective is our method at combating style copying by text-to-video AI generators?
3. What limitations are there to our method and what future research directions are there?

Value to User Community

Prospective users of our project are artists and animators who want to combat the rise in generative AI trained on their work. Although style protection measures exist for singular artworks (photos, illustrations, etc.), there are currently no measures to prevent style copying for videos (such as animated gifs or films). Our project fills this void by designing software that can be used for this purpose. We will make our work available to them through a public github repository that can be downloaded and applied similarly to how standard Glaze can be downloaded and applied.

Demo

1-minute elevator pitch

Our elevator pitch will involve explaining:

1. What we mean by style copying with a photo/illustration example (this will largely follow Glaze's own explanations)
2. Glaze and what it does
3. How we've adapted/applied Glaze to videos and animations
4. How effective our application was at preventing style copying (what text-to-video generators we tested it on and how we rate their ability to copy the animations after they've been video Glazed by our project)

5-minute demo:

Our demo will expand on the elevator pitch, using more examples. Since video-Glaze takes some time to run, we will run the animations through it ahead of time and focus the demo on showing how this prevents style-copying. After explaining what Glaze is, how it works, what its purpose is, and generally what our adaptation of it is, we will choose an animation to demonstrate the video-Glazing process. First, we'll show the un-Glazed video, then show the Glazed one (that we've already run through our software ahead of time) and emphasize how the naked eye can't see much difference between the two videos. Then we'll run both the un-Glazed and Glazed videos through one of our chosen text-to-video AI generators with a prompt to make a similar animation in the same style, demonstrating how the generator succeeds with the un-Glazed video but fails with the Glazed one. Then we'll briefly go over our studies of all our animation clips, explaining when our method is more effective and when it doesn't work as well.

Delivery

We will deliver a [github repository](#) with our video-modified Glaze software, our [animations](#), and our testing software.

Our Progress and Next Steps

We have achieved the following already:

1. Compiled animation clips (from recent open source data set)
 - a. Find animation clips from a variety of years/popularity so that we can compare if text-to-video generators are effective at copying glazed videos if they've been seen before/probably used in the training data
 - b. Choose short clips (~50 frames), gif-length
 - c. Select 7 clips
2. Write a script for applying glaze frame by frame

We plan to achieve the following by the end of the semester:

1. Apply glaze frame-by-frame to all clips
2. Submit each glazed animation to the 3 text-to-video generators
3. Come up with more detailed and concrete evaluation metrics
 - a. Human - survey 10-20 people (could post to class on Ed potentially)
 - b. Frame-level-similarity
 - c. Some sort of style mimicry
4. Actually evaluate the videos based on the theorized evaluation criteria
5. Compare literature review to the results we found
6. Test our algorithm on more clips/larger clips if needed

References

- [1] S. Shan, J. Cryan, E. Wenger, H. Zheng, R. Hanocka, and B. Y. Zhao, "Glaze: Protecting artists from style mimicry by Text-to-Image models," in 32nd USENIX Security Symposium (USENIX Security 23). Anaheim, CA: USENIX Association, Aug. 2023, pp. 2187–2204. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity23/presentation/shan>
- [2] C. Murray, "Controversial AI 'Actress' Tilly Norwood Rebuffs Backlash In New Song," *Forbes*, Mar. 10, 2026. Available: <https://www.forbes.com/sites/conormurray/2026/03/10/ai-actress-tilly-norwood-responds-to-backlash-in-debut-song-ais-not-the-enemy/>
- [3] Szarowicz, A., Amiguet-Vercher, J., Forte, P., Briggs, J., Gelepithis, P., Remagnino, P. (2001). The Application of AI to Automatically Generated Animation. In: Stumptner, M., Corbett, D., Brooks, M. (eds) AI 2001: Advances in Artificial Intelligence. AI 2001. Lecture Notes in Computer Science(), vol 2256. Springer, Berlin, Heidelberg. https://doi.org/10.1007/3-540-45656-2_42.
- [4] E. Guo, P. Li, S. Yu and H. Wang, "Efficient Video Privacy Protection Against Malicious Face Recognition Models," in *IEEE Open Journal of the Computer Society*, vol. 3, pp. 271-280, 2022, doi: 10.1109/OJCS.2022.3218559. Url = <https://ieeexplore.ieee.org/abstract/document/9933823>.
- [5] Z. Guo, J. Xu and J. Xu, "Protecting Personal Privacy Against Unauthorized Facial Recognition Models via Image Cloaking," *2025 11th IEEE International Conference on Privacy Computing and Data Security (PCDS)*, Hakodate, Japan, 2025, pp. 57-64, doi: 10.1109/PCDS65695.2025.00017. <https://ieeexplore.ieee.org/abstract/document/11172832>
- [6] Benoit, A., Ciobotaru, M., Lambert, P., Ionescu, B. (2011). Similarity Measurement for Animation Movies. In: Lee, KT., Tsai, WH., Liao, HY.M., Chen, T., Hsieh, JW., Tseng, CC. (eds) Advances in Multimedia Modeling. MMM 2011. Lecture Notes in Computer Science, vol 6523. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-17832-0_33.
- [7] S. S. Cheung and A. Zakhor, "Efficient video similarity measurement with video signature," *Proceedings. International Conference on Image Processing*, Rochester, NY, USA, 2002, pp. I-I, doi: 10.1109/ICIP.2002.1038101. <https://ieeexplore.ieee.org/abstract/document/1038101>